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Sustainable Utilization of Zero Liquid Discharge Produced Water Treatment Waste in Lost Circulation Applications

J. Ramasamy, M. K. Al-Arfaj, S. C. Ayirala, and S. H. Saleh, Drilling Technology Division, Exploration and Petroleum Engineering Advanced Research Center, Saudi Aramco, Dhahran, Saudi Arabia

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Abstract

In the process of oil and gas explorations, crude is being produced from subsurface reservoirs. Crude is a mixture of hydrocarbon resources such as oil and gas along with water. The mixture is then sent to gas oil separating plant (GOSP) where oil, gas, water and other impurities get removed and/or separated. Suitable treatment and handling technologies of produced water for proper recycling and reuse is gaining traction within oil and gas industries. Zero liquid discharge (ZLD) technology developed exclusively for desalination of produced water generates low saline clean water from hypersaline produced water with recovery efficiency from 80 to 90%. The small quantity of excessively salt laden reject waste stream is generated as a byproduct from ZLD desalination plant. As the waste water is saturated with salts, it is not fit to reuse for many domains unless treated appropriately. Finding use for reject waste water to reuse without further treatment such as recovery of different minerals or any drilling applications is ideal to accomplish 100% circular economy. In this paper, we studied the applicability of waste water for potential reuse in drilling applications as carrier fluid to prepare lost circulation slurry and compared the performance with lost circulation slurry prepared using regular drill water. We have also investigated the applicability of ZLD reject brine as an internal phase of formulating invert emulsion oil based mud. This is to promote sustainability to reuse the waste water and to circumvent waste discarding.

Introduction

Hydrocarbon energy has been an inseparable part for continuous economic growth in the world. Oil and gas bearing reservoir rocks in the subsurface also contain water. The water also is produced to surface along with oil and gas during production of hydrocarbon (Healy et al. 2015). This water is called as produced water and obtained as a byproduct. The produced water is separated in GOSP from oil and gas. The quantity of produced water will vary from reservoir to reservoir. The quantity and composition of produced water changes during the lifetime of the well (Nasiri et al. 2017). Proper treatment, reuse and/or removal of produced water is critical in terms of sustainability and protection of the environment. The geochemistry of produced water depends on the reservoir and vary extensively in salinity. In some instances, the salinity of produced water can be several folds higher than that of sea water (Nasiri et al. 2017). Produced water usually contains traces of oil, mud, residues, sand or mud, chemicals from frac fluids, bacteria, and dissolved organic compounds (Nasiri et al. 2017). There are several methods reported for treating produced water (Amakiri et al. 2022) and some of the technologies are environment benign with low carbon footprint (Aljedaani et al. 2021, Ayirala et al. 2018). The plant designed for desalination of produced water is also known as Zero Liquid Discharge (ZLD) technology which employs a pretreatment system to lower residual oil and dissolved H_2S content to less than 10 ppm in produced water. The pretreated produced water is then subjected to an evaporation based "dynamic vapor recovery" technology to desalinate the produced water. The treated clean water obtained from desalination of produced water can be utilized for various upstream applications such as injection water for water flooding, makeup water for different improved/enhanced oil recovery (IOR/EOR) processes, for fracking operations, wash water for crude oil desalting, and potentially as a water resource for irrigation. Even though the recovery efficiencies of ZLD produced water desalination system are up to 90%, it generates up to 10% volume of hyper saturated and concentrated brine enriched with inorganic salts discharged as waste. Discarding the concentrated waste brines into the sea will have an impact on the organisms in and around the area of discharge. Discharge on land on the other hand, will leave deposition of salt residue after evaporation. Hyper salinity nature of the discharge water makes it hard to identify recycling options. Appropriate reuse of waste water discharge is important to realize 100% circular economy and contribute to environmental sustainability.

The upstream process of oil and gas extraction involves circulation of drilling mud throughout the entire time of drilling. Subsurface formation has varying degrees of porosity and permeability. There could be tight rock formations with very little permeability. There are formations with high porous and permeable characteristics. In some formations, there are fractures present naturally and while drilling through such formations, loss of circulation will occur. Fractures could also be induced in weak formation due to improper mud weight and high ECD. Loss of circulation will occur in such scenarios as well. In order to plug the fractures and control the loss of fluid into the formation, a pill containing lost circulation material (LCM) is pumped from the surface. LCM is usually a mixture of particulate, fibrous, flaky and dissolvable materials that form a bridge across the fracture to reduce fluid invasion. Regular drill water is typically used as a carrier fluid for LCM pill. The source of the water to formulate water based mud and LCM pill is obtained from water well. We have already reported the application of ZLD reject water in preparing regular water based drilling fluid by having 50:50 ratio of regular water and ZLD waste water (Ramasamy et al. 2024). Using Possibility of cross contamination of hypersaline brine reject water with drilling fluid and cause hampering of drilling fluid properties is very less as the LCM pill usually be inside the fractures. Any residual LCM pill in the wellbore after curing losses will be removed by circulating a Hi-Vis pill to remove the residual LCM pill back to the surface. Therefore, it is appropriate to use ZLD waste reject water for LCM pill preparation applications as the waste water is sent back to subsurface from where it was produced originally. However, appropriate testing and screening of the waste water and make them suitable for applications in LCM pill preparation will create significant value for circular economy and sustainability for the waste water obtained from various operational streams. As produced water is originally extracted with oil and gas in crude, it is appropriate to identify applications for the waste generated from produced water in one of the oil and gas operations. The sequence of process starting form crude to waste water is shown figure 1. Several reports have been published towards sustainable and environmentally friendly technologies for oil and gas drilling operations for lost circulation mitigation (Ramasamy et al. 2017, Ramasamy et al. 2018), additives for water based (Ramasamy et al. 2022) and oil based mud additives (Ramasamy et al. 2019, Ramasamy et al. 2019). In this paper, we have shown the applicability of ZLD waste water discharge for LCM pill preparation in oil and gas industry drilling process. It offers a win-win approach on two different fronts as it enables recycling of waste water to avoid disposal concerns for the environment and significantly reduce fresh water volumes used in drilling operations.

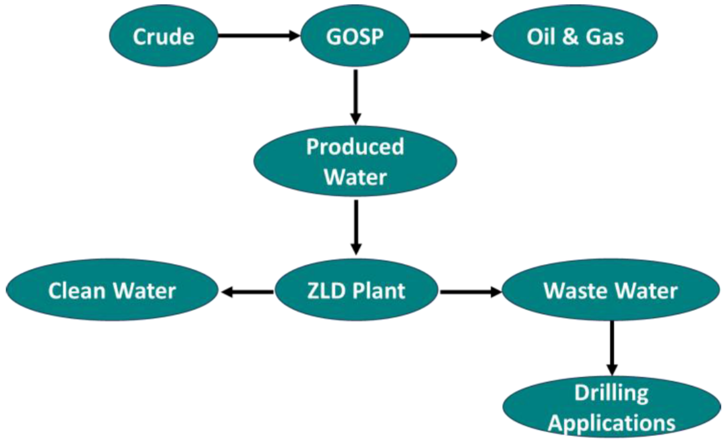

Figure 1—Recycling process of produced waste water

ZLD Waste Water

The ZLD waste water reject stream is extremely salt laden. The salinity of waste water is several folds higher than sea water. Different ions present in the waste reject water is given in Table 1. Total dissolved solids (TDS) is in the range of 360000 ppm with largely contains sodium and chloride ions along with other mono valent and divalent ions.

Table 1—Ionic profile of waste water discharge

Parameter	ppm	Concentration
Total Dissolved Solids	mg/L	360000
Sodium Ions	mg/l	143204
Potassium Ions	mg/l	4902
Calcium Ions	mg/L	36719
Magnesium Ions	mg/L	4149
Sulphate Ions	mg/L	2269
Chloride Ions	mg/L	166451
Bicarbonate Ions	mg/L	2306

Results and Discussion

Typical loss circulation pill formulation used to mitigate partial loss of circulation is given in table 2. Pill #1 is formulated using regular water while Pill #2 is prepared using ZLD reject waste water as carrier fluid. In the two formulations, xanthan gum is used as viscosifier as routinely used viscosifier bentonite will not work due to excessive hardness in the waste water and it wouldn't yield to provide viscosity.

Table 2—Loss circulation pill formulation using regular water and plant waste water

Products	Pill #1	Pill #2
	Carrier Fluid	
Regular water	0.72 BBL	
Wastewater		0.72 BBL
Xanthan Gum	1.5 PPB	1.5 PPB
	LCM Slurry (PPB)	
Particulate LCM (ARC Plug)	15	15
Fibrous LCM Coarse (ARC Eco-Fiber C)	10	10
Marble Coarse	20	20
Marble Medium	20	20
Calcium Carbonate Fine	10	10
Graphite Fine	10	10
Marble Flake Medium	15	15
Fibrous LCM Medium	10	10
Fibrous LCM Fine	3	3

The LCM pill contains regular loss circulation prevention additives such as particulate, flaky and fibrous materials along with acid soluble additives. Laboratory LCM screening is done using the industry standard LCM testing device shown in figure 2. The equipment offers screening ability at varying temperature and pressure conditions in order to simulate downhole conditions. Slotted disks are used to simulate fractured formation and the tests will identify the ability of the LCM to plug the slots. There are different slotted disks with varying widths are available. In our study, we have used 0.5 mm width slotted disk that represents partial loss of circulation.

Figure 2—Loss circulation material screening device

Table 3 shows the LCM test results conducted for pill #1 and pill #2. The tests were conducted at 1500 psi. Temperature screening was also done using room temperature and at elevated temperature of 250 #F for pill #2. There is no loss or negligible loss is observed for all three screening tests which is typical for the pill recipe. The important observation to note is there is no difference in terms of pill suspension, stability and performance. The hyper saline ZLD waste water based pill #2 has performed as good as the pill #1 which is formulated using regular water. The salinity did not have any detrimental effect either on the pill stability and performance.

Table 3—Lost circulation material testing

Pill #	Slot width	Temperature	Pressure	Fluid Loss
Pill #1	0.5 mm	Room Temp	1500 psi	No/negligible loss
Pill #2	0.5 mm	Room Temp	1500 psi	No/negligible loss
Pill #2	0.5 mm	250 °F	1500 psi	No/negligible loss

Oil based mud is used in drilling for several reasons mainly in drilling zones where water based mud will fail to perform due to conditions exist in the downhole. One of the main reasons why oil based mud is used because it has better thermal stability than water based mud. on-reactive nature of oil based mud with shale section makes it suitable to drill through highly reactive shale formation. Moreover, inherent lubricious property of oil based mud will make it suitable mud system for drilling long horizontal and extended reach wells. Oil based mud is widely used in the form of invert emulsion oil based mud where water in used as an internal phase while oil is a continuous phase to prepare invert emulsion. Water is used in oil based mud for various reasons mainly to increase the density of oil based mud by using different brine system such as CaCl2, NaCl, KCl, etc. Another reason to use brine in the oil based mud is to stabilize shale section by reverse osmosis process in which water from shale section is migrated to water in phase in oil due to difference in salt concentration. The invert emulsion oil based mud has been successfully used in oil and gas drilling for several decades.

Zero liquid discharge waste reject water is a concentrated brine consisting of several salt combinations including mono and divalent ions. Instead of using freshly prepared brine, it may be appropriate to use ZLD reject water as it is enriched with salts in invert emulsion oil based mud to use across zones that are not challenging which may require extreme care. The additives and polymers such as primary and secondary emulsifiers which are derived from fatty acid derivatives, organophilic lignite, and organophilic clay need to be compatible with hyper saline waste water in the internal phase. We have formulated a typical invert emulsion oil based mud using regular additives. Four invert emulsion oil based fluid samples were formulated in order to screen and evaluate the applicability of concentrated brine waste water as an internal phase. Regularly used oil based mud chemicals were utilized to prepare the formulations. For two formulations OBM-1 and OBM-2, diesel is used as base or continuous. The other two formulations OBM-3 and OBM-4 were formulated using Safra oil which is a low toxicity mineral oil as base or continuous fluid. In each type of base fluid, formulations with 100% regular water based brine and 50:50 regular water and concentrated waste based brine were used as internal phase. The formulations were given in the table 4 and 5. There is no issue observed like foaming and separation while mixing the mud. The additives mixed well in the mud during preparation.

Table 4—Diesel based formulations

Additives	OBM-1	OBM-2
Diesel (ml)	218	218
Primary Emulsifier (ml)	8	8
Secondary Emulsifier (ml)	4	4
Lime (g)	6	6
Viscosifier (g)	4	4
Fluid loss control additive (g)	6	6
Brine with 61 g CaCl2 (waste brine : regular water)	85 (100%) regular water only	85 (50/50)%
Barite (g)	161	161

Table 5—Safra oil-based formulations

Additives	OBM-3	OBM-4
Safra Oil (ml)	218	218
Primary Emulsifier (ml)	8	8
Secondary Emulsifier (ml)	4	4
Lime (g)	6	6
Viscosifier (g)	4	4
Fluid loss control additive (g)	6	6
Brine with 61 g CaCl2 (waste brine: regular water)	85 (100%) regular water only	85 (50/50)%
Barite (g)	161	161

The fluids were subjected to downhole conditions by hot rolling for 16 h at of 300 #F and 500 psi. the Rheological properties of OBM-1 is similar to that of OBM-2 which has concentrated waste as brine along with tap water as shown in table 6 as can be seen from yield point (YP) values. The plastic viscosity (PV) is also better for OBM-2 as compared with OBM-1. The fluid loss control tests were done at 300 #F and 500 psi differential pressure. OBM-2 has slightly better fluid loss control property as compared to OBM-1.

Table 6—Mud properties for Diesel based mud.

Mud Properties	OBM-1	OBM-2
PV	15	12
YP	3	4
HTHP Spurt Loss	2	4
HTHP Total Loss	14	6
HTHP Mud Cake (mm)	3.175	4.175

Similar to diesel-based mud formulations, the safra oil based OBM-3 and OBM-4 were also tested for rheological and fluid loss control properties as shown in table 7. As observed in the case of diesel-based fluid, similar rheological performance is observed for OBM-3 and OBM-4. On the other hand, better fluid loss control is achieved for the mud having concentrated waste brine in the internal phase.

Table 7—Mud properties for Safra oil based mud.

Mud Properties	OBM-3	OBM-4
PV	13	23
YP	13	13
HTHP Spurt Loss	6	2
HTHP Total Loss	10	6
HTHP Mud Cake (mm)	4.175	3.175

It is evident from the results discussed above, that the ZLD concentrated waste brine discharged from produced water desalination processes could be used as an internal phase of invert emulsion oil-based mud without compromising any required mud properties such as rheological and fluid loss control. Moreover, better emulsion stability is also observed in the filtrate collected from HPHT fluid loss control experiment. There was no phase separation of oil and water in the filtrate observed even after 24 h. Therefore, the equal mixture of concentrated waste brine and the regular water as 50:50 composition can be used to prepare invert-emulsion oil-based mud systems.

Conclusion

In conclusion, we have tested and identified potential usage of salt laden ZLD waste water discharge from desalination plant as a carrier fluid to formulate loss circulation control pill and as an internal phase in invert emulsion oil based mud systems. There is no issue observed while mixing of the mud samples. The outcome of the performance tests also revealed that the samples were performed similar or better than the mud samples prepared using regular water. This demonstrates the potential application of highly salt concentrated waste reject streams obtained from ZLD produced water management solution in drilling operations to prepare mud systems and loss circulation mitigation pill formulations.to formulate water-based mud formulations for drilling applications to achieve circular economy and environmental sustainability.

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